

**A DATA DRIVEN INVESTIGATION INTO STUDENT
DEPRESSION: PREDICTION MODEL AND INFERENTIAL
APPROACHES**

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SUPERVISOR CERTIFICATE

This is certified that the project paper entitled “ **A Data Driven Investigation Into Student Depression: Prediction Model and Inferential Approaches**” submitted by Ananya Paul in partial fulfillment of the requirement for the Bachelor degree of Statistics (Honours) is based upon the result of benefited research work carried out by the investigator under my guidance and supervision.

The results of the investigator reported in this project paper have not so far been submitted for any degree or diploma.

Signature

DECLARATION

I am **Ananya Paul**, a student of **B.Sc. Semester 6, Statistics Honours**, of **University of Calcutta**, Registration no. **012-1211-0624-22**, Roll No. **223012-11-0031**, hereby declare that this project report, **“A Data Driven Investigation Into Student Depression: Prediction Model and Inferential Approaches”** is my own work under the supervision of **Oindrila Bose Ma’am** as a part of B.Sc Sem 6 Examination (Paper DSE-B2) and that all sources of information and data used have been acknowledged.

I affirm that this **work has not been copied or plagiarized** from any other source and that it has not been submitted previously for any academic award or examination.

Signature:

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ABSTRACT

Our study aimed to **investigate the key demographic, academic, lifestyle, and psychosocial factors associated with depression among students using a comprehensive statistical framework.** The dataset analyzed comprised variables such as **age, gender, CGPA, academic pressure, study satisfaction, sleep duration, dietary habits, work-study hours, financial stress, family history of mental illness, and suicidal thoughts.** Preliminary exploratory analyses were conducted through **boxplots, stacked bar plots, and correlation heatmaps** to gain initial insights into data distribution and inter-variable relationships.

To formally **assess associations, chi-square tests of independence were performed for categorical variables,** revealing significant relationships between depression and factors such as academic pressure, financial stress, family history of mental illness, and suicidal ideation. Given the **non-normal distribution of CGPA across depression groups confirmed through QQ plots,** the **Wilcoxon rank-sum test was utilized to compare medians.** Additionally, a **binomial proportion test was conducted to evaluate differences in depression prevalence across three distinct age groups, highlighting younger students as more vulnerable.**

The core of the analysis involved fitting a **multiple logistic regression model** to quantify the influence of eleven predictor variables on the **likelihood of experiencing depression. Dummy variables were created for multi-level categorical predictors, with reference categories carefully chosen to facilitate interpretation.** The model exhibited strong predictive capability, as evidenced by a high **AUC value** of approximately 0.92 from **ROC analysis,** alongside favourable sensitivity and specificity outcomes. Significant predictors included **high levels of academic pressure, severe financial stress, unhealthy dietary patterns, reduced sleep duration, a family history of mental illness, and suicidal thoughts.**

The scope of this project extends beyond mere statistical modeling to provide actionable insights into the multifactorial nature of student depression. By identifying the most impactful risk factors, the study lays the **groundwork for targeted interventions and campus wellness programs aimed at mitigating mental health challenges among students.** These findings not only enhance our understanding of the determinants of student depression but also inform stakeholders in educational and healthcare settings to prioritize mental health initiatives tailored to student needs.

INTRODUCTION

Depression is a pervasive mental health disorder that affects individuals across all age groups, with university and college students emerging as an especially vulnerable population. The academic environment often imposes multiple stressors — from rigorous coursework and examinations to uncertainties about future careers — which, compounded by personal and socio-economic challenges, can substantially elevate the risk of psychological distress. **In recent years, concerns over student mental health have intensified globally, underscoring the need for data-driven approaches to identify key determinants and facilitate timely interventions.**

The **multifaceted nature of depression means that it seldom arises from a single cause**. Instead, it is influenced by an interplay of demographic factors (such as age and gender), academic pressures (like high workloads and low satisfaction with studies), lifestyle habits (including diet and sleep), financial concerns, and family history of mental illness. Recognizing these diverse contributors is critical for developing comprehensive mental health strategies within educational institutions.

The primary objective of this study is to statistically examine and quantify the factors associated with depression among students. Using a rich dataset encompassing a variety of academic, lifestyle, and psychosocial variables, the study seeks to identify which factors most significantly increase the likelihood of depression. By employing a combination of exploratory data analysis, hypothesis testing (including chi-square tests, non-parametric comparisons, and proportion tests), and advanced modeling through logistic regression, the project aims to provide a holistic understanding of the drivers of student depression.

METHODOLOGY

The present study aims to identify and **quantify the factors** associated with depression among students using statistical modelling techniques. To achieve this, we employed a comprehensive methodology comprising **data preparation, exploratory analysis, hypothesis testing, and logistic regression modelling**. This systematic approach enabled us to examine the relationships between various predictor variables—such as academic pressure, sleep duration, dietary habits, and financial stress—and the likelihood of experiencing depression.

The methodology used for different sections of this report is given below:

- **DATA PREPARATION:**

- a) **Data cleaning:** The data was checked for **missing** and invalid values, which were removed using **na.omit function in R**. Moreover the unimportant columns were removed.
- b) **Variable encoding:** Categorical variables such as *Gender, Academic Pressure, Study Satisfaction, Sleep Duration, Dietary Habits, Financial Stress, Family History of Mental Illness, and Have you ever had suicidal thoughts* were converted into **factors** with meaningful labels using **as.factor function in R**. Continuous variables like *Age, CGPA, and Work-Study Hours* were retained in **numeric** form.

- **DATA VISUALISATION:**

- a) **Boxplots :** A boxplot is a graphical representation of the distribution of a dataset that shows the median, quartiles, and potential outliers, providing a summary of central tendency and variability. Boxplot of *CGPA* and *Sleep Duration* stratified by depression status were generated to examine differences in **central tendency, spread, and the presence of outliers between depressed and non-depressed students**. These visualizations using the **boxplot() function in R**
- b) **Stackplots :** A stacked bar plot displays the composition of categorical data across groups by stacking segments within bars, allowing comparison of relative frequencies or proportions. To investigate the distribution of categorical predictors across depression status, stacked bar plots were utilized. These plots provided a clear visual representation of how **proportions of categories** such as *Academic Pressure, Financial Stress, Study Satisfaction, and Suicidal Thoughts* **varied between**

depressed and non-depressed students. The stacked bar plots were generated using the **barplot()** function in R.

- c) **Correlation heatmap:** A heatmap is a **color-coded matrix** that visually represents the magnitude of relationships or values, commonly used to display correlation coefficients between variables. To assess the strength and direction of linear relationships among continuous and ordinal variables in the dataset, a correlation heatmap was constructed. The correlation matrix was computed using the **cor()** function in R, which **quantified pairwise Pearson correlation coefficients.**

- **TESTING OF HYPOTHESES:**

- a) **Chi Square tests:** To statistically examine the association between depression status and various categorical predictor variables, chi-square tests of independence were performed. **Contingency tables** were first created using the **table()** function in R to summarize the joint frequencies of depression status with factors such as *Academic Pressure*, *Study Satisfaction*, *Financial Stress*, *Family History of Mental Illness*, and *Suicidal Thoughts*. The **chisq.test()** function was then applied to these tables to compute the **chi-square test statistic, degrees of freedom, and p-values.** For 2×2 tables, **Yates' continuity correction was employed by default to adjust for potential overestimation of significance.** A significance level of $\alpha = 0.05$ was used to determine whether the observed associations were statistically significant. This procedure allowed us to formally test the null hypothesis that depression status is independent of each categorical predictor variable.
- b) **Wilcoxon test:** To assess whether the continuous variable *CGPA* was normally distributed within groups defined by depression status, **quantile-quantile (QQ) plots** were generated using the **qqnorm()** and **qqline()** functions in R. Visual inspection of these plots indicated notable departures from normality, leading to the selection of a non-parametric test for comparing groups. Accordingly, the **Wilcoxon rank-sum test (Mann-Whitney U test)** was employed using the **wilcox.test()**

function to evaluate whether the distributions of *CGPA* differed between depressed and non-depressed students.

- c) **Test for equality of proportion:** To investigate whether the proportion of depressed students differed significantly across age groups, a binomial **proportion test** was performed. The continuous *Age* variable was first categorized into three groups: **≤20 years, 21–23 years, and >23 years**. Within each group, the proportion of students classified as depressed was calculated. To formally test whether these proportions were equal across the age categories, the **prop.test()** function in R was utilized, which conducts a chi-square test for equality of proportions based on binomial counts

LOGISTIC REGRESSION

The logistic regression is the linear combination of predictors. And it models the **log odds of p** with respect to different predictors.

- A **multiple logistic regression model** was employed to investigate the relationship between depression status (binary outcome: depressed vs. non-depressed) and a set of explanatory variables.
- The model incorporated **eleven predictors**:
 - *Gender, Age, Academic Pressure, CGPA, Study Satisfaction, Sleep Duration, Dietary Habits, Work-Study Hours, Financial Stress, Family History of Mental Illness, and Suicidal Thoughts.*
- **Categorical predictors were converted into sets of dummy variables (indicator variables)**, with one level treated as the **reference category**, to allow interpretation of effects relative to the baseline.
- The logistic regression was fit using the **glm()** function in R with the **binomial family and logit link**.
- Model parameters were estimated by **maximum likelihood estimation (via Fisher scoring)**.
- **Goodness-of-fit and predictive performance** were assessed through construction of a **confusion matrix**, calculation of **sensitivity and specificity**, and generation of a **ROC curve** along with computation of the **area under the curve (AUC)** to quantify the model's classification ability.

DATA DESCRIPTION

The **STUDENT DEPRESSION DATASET** includes data for predicting depression among students from different parts of India using several medical and psychological parameters that allegedly affect the mental health of students under study.

The dataset has **27901 rows and 18 columns** before preprocessing that can work as predictors in our analysis. The variables are described below:

VARIABLE	DESCRIPTION	TYPE
Id	Unique Identifier for each student.	Numeric
Gender	Gender of the student	Categorical
Age	Age of the student in years	Numeric
City	Residence of the student	Categorical
Profession	Occupation of student	Categorical
Academic Pressure	Self rated academic stress level	Ordinal
Work Pressure	Self rated work stress level	Ordinal
CGPA	Cumulative Grade POINT Average	Numeric
Study Satisfaction	Satisfaction with academic environment	Ordinal
Job Satisfaction	Satisfaction with work (if employed)	Ordinal
Sleep Duration	Average nightly sleep	Categorical
Dietary Habit	Type of diet	Categorical
Degree	Educational Qualification Pursued	Categorical
Have you ever had suicidal thoughts	Whether the student has ever had suicidal ideation	Binary
Work/Study hours	Number of hours spent daily on work or study	Numeric
Financial Stress	Self-reported financial pressure	Ordinal
Family history of mental illness	Whether any family member had mental illness	Binary
Depression	Whether the student is classified as depressed	Binary (target variable)

DATA CLEANING AND PREPROCESSING

The dataset, comprising responses from 27,901 students on factors related to depression, underwent a **systematic cleaning process to ensure its quality and reliability for subsequent statistical analysis**.

The dataset was notably complete, with only **three missing entries** in the **FINANCIAL STRESS** variable. These missing values were removed.

Further all the categorical variables eg. Depression status, Sleep duration, Dietary habits, Gender, Financial stress, Family history of mental illness, Pressure level, Study hours, Satisfaction level to **numerical codes**. This facilitated easier interpretation and computational processing.

Next, in the dataset we notice that profession of all individuals is “Student”, so it will not affect our analysis if removed. Further the Work Pressure column and Job Satisfaction column have all values as 0. The Degree pursued by students has no important role in our analysis and Id of students is a unique column. So we delete these columns from the dataset.

Post cleaning, we have 27898 observations from 12 variables. The variables and their type are as follows:

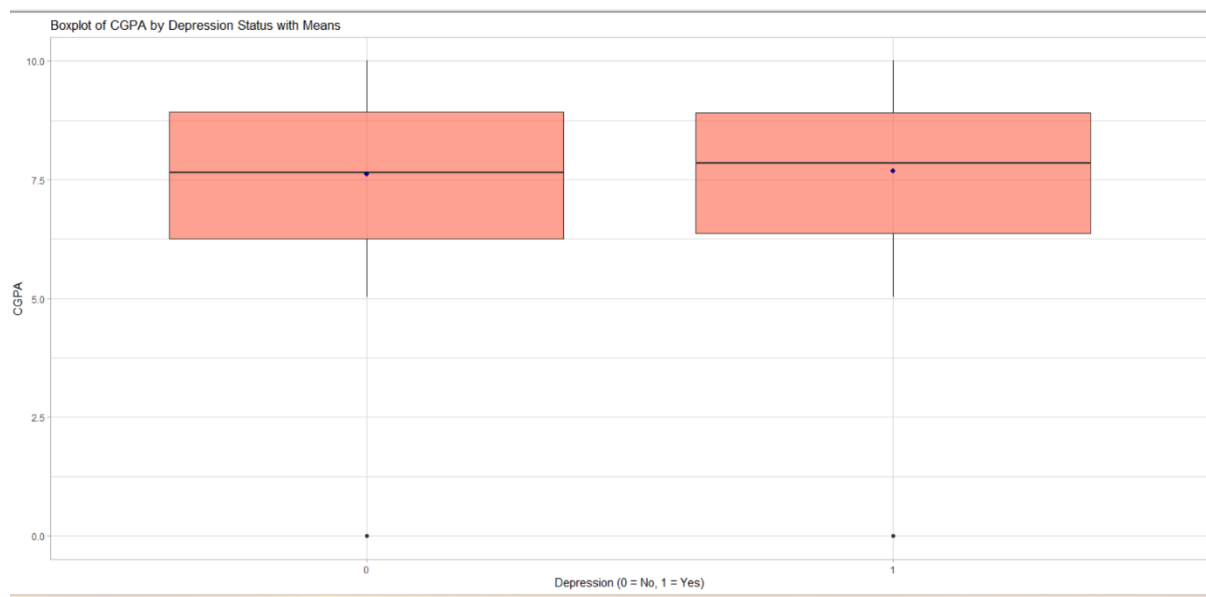
```
> str(data1)
'data.frame':  27898 obs. of  12 variables:
 $ Gender          : Factor w/ 2 levels "Female","Male": 2 1 2 1 1 2 2 1 2 2 ...
 $ Age             : num  33 24 31 28 25 29 30 30 28 31 ...
 $ Academic.Pressure : Factor w/ 6 levels "0","1","2","3",...: 6 3 4 4 5 3 4 3 4 3 ...
 $ CGPA            : num  8.97 5.9 7.03 5.59 8.13 5.7 9.54 8.04 9.79 8.38 ...
 $ Study.Satisfaction : Factor w/ 6 levels "0","1","2","3",...: 3 6 6 3 4 4 5 5 2 4 ...
 $ Sleep.Duration   : Factor w/ 5 levels "5-6 hours","7-8 hours",...: 1 1 3 2 1 3 2 3 2 3 ...
 $ Dietary.Habits    : Factor w/ 4 levels "Healthy","Moderate",...: 1 2 1 2 2 1 1 4 2 2 ...
 $ Work.Study.Hours  : num  3 3 9 4 1 4 1 0 12 2 ...
 $ Financial.Stress  : Factor w/ 5 levels "1","2","3","4",...: 1 2 1 5 1 1 2 1 3 5 ...
 $ Family.History.of.Mental.Illness : Factor w/ 2 levels "No","Yes": 1 2 2 2 1 1 1 2 1 1 ...
 $ Depression        : Factor w/ 2 levels "0","1": 2 1 1 2 1 1 1 1 2 2 ...
 $ Have.you.ever.had.suicidal.thoughts: Factor w/ 2 levels "No","Yes": 2 1 1 2 2 1 1 1 2 2 ...
```

After data preprocessing, we shall now proceed to **Data Visualisation**.

DATA VISUALISATION

To gain an initial understanding of the dataset and **uncover potential patterns** related to student depression, **various graphical techniques** were employed. These visual explorations provided a foundational perspective for subsequent statistical analyses and model development.

1. BOXPLOT OF CGPA VS DEPRESSION STATUS



INTERPRETATION OF THE BOXPLOT:

X-axis: Depression (0 = No, 1 = Yes)

0 means students not depressed

1 means students depressed

Y-axis: CGPA (ranges roughly from 0 to 10)

Boxplots: show the distribution of CGPA scores for each group.

Median comparison:

The **thick black line** inside each box represents the median CGPA.

Here, the median for: Non-depressed (0) students is slightly lower than for depressed (1) students.

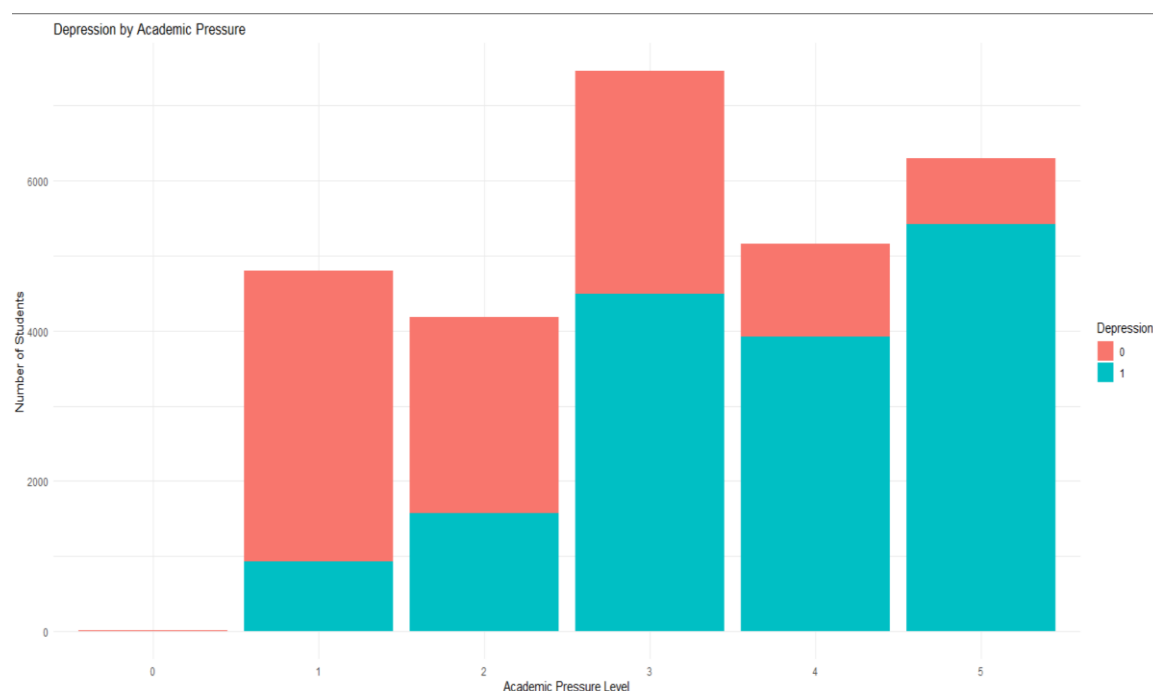
Outlier detection:

Tiny dots near the bottom — these are outliers (unusually low CGPAs).

There's an extreme low outlier near 0 CGPA in both groups.

Comment: Visually, students with depression tend to have slightly higher CGPAs on average than students without depression. The boxplot reveals that while students with depression tend to have slightly higher median and mean CGPAs compared to those without depression, the distributions show substantial overlap. This suggests that CGPA alone does not strongly differentiate between depressed and non-depressed students. Formal statistical testing (like a t-test) is required to determine if the observed differences are significant.

2. STACKED BAR PLOT OF ACADEMIC PRESSURE AGAINST DEPRESSION STATUS



INTERPRETATION OF THE STACKBAR

X-axis: Academic Pressure Level (from 0 to 5).

0 = No pressure, 5 = Extreme pressure.

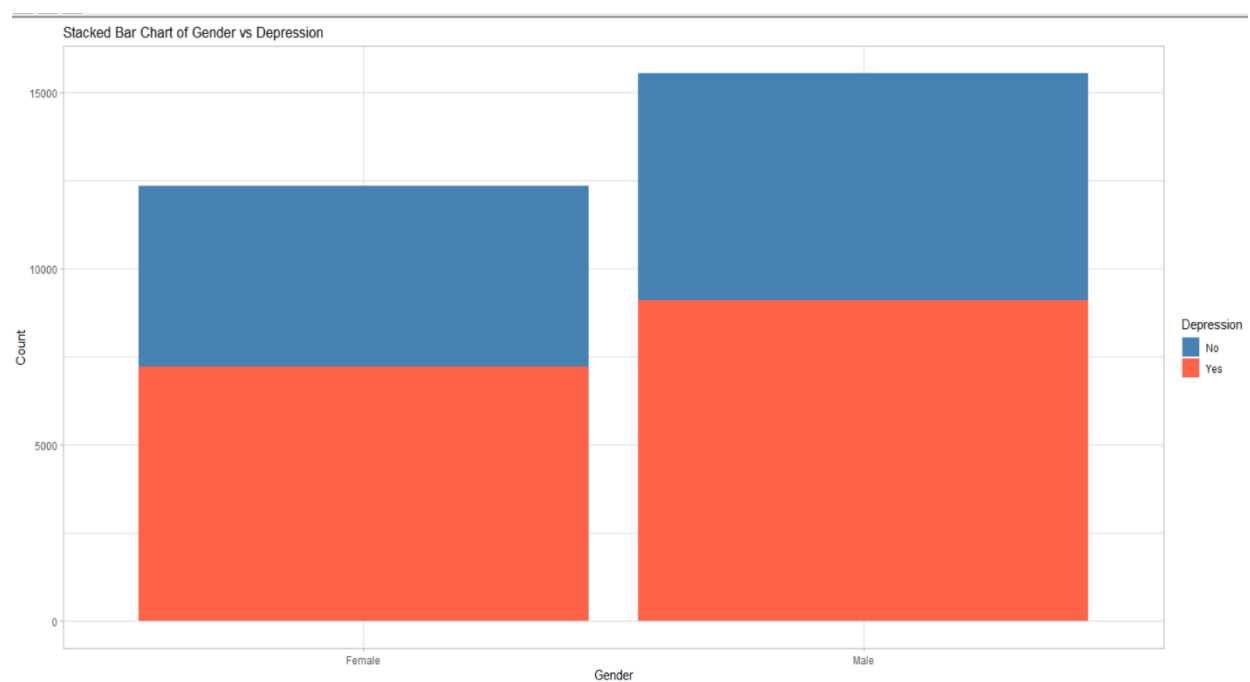
Y-axis: Number of Students in each category.

Proportion depressed at each level:

- Academic Pressure **Level 1 (Very Low)**: Has a small majority of depressed students (blue) compared to non-depressed.
- Academic Pressure **Levels 2 and 3 (Moderate)**: At level 3, the numbers peak and this group has the largest number of students overall. Majority at level 3 are not depressed (red), though a sizeable portion still are.
- Academic Pressure **Levels 4 and 5 (High to Very High)**: At level 5, most students are depressed (blue dominates).

Comment: The stacked bar plot illustrates that while students report depression across all levels of academic pressure, the proportion of depressed students increases notably at the highest pressure level (5). This suggests a **potential positive relationship between academic stress and depression**. However, the presence of substantial depression even at low pressure levels indicates that other factors beyond academic pressure contribute to depression. A Chi-square test would statistically test the association.

3. STACKBAR PLOT OF GENDER AGAINST DEPRESSION STATUS



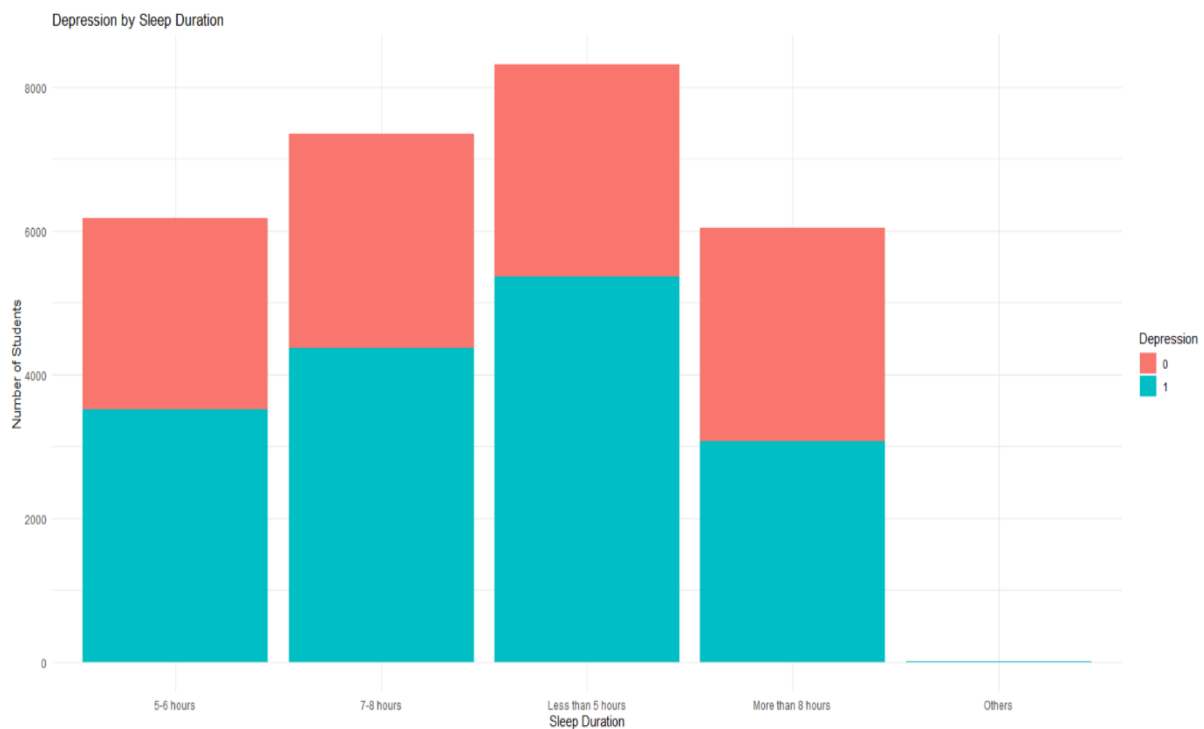
INTERPRETATION OF STACKBAR

It is clear from the heights of the bars that: There are more male students than female students in this dataset. (The male bar is visibly taller overall.)

The absolute number of depressed males is higher than that of females, primarily because there are more male students overall in the dataset. However, if we look at the relative split inside each bar, **the proportion of depressed students is fairly similar across genders.**

This suggests **no pronounced gender disparity in the prevalence of depression on a relative scale.** Formal testing using a Chi-square test for independence could further quantify this observation.

4. STACKBAR PLOT OF SLEEP DURATION VS DEPRESSION



INTERPRETATION OF THE STACKBAR

Proportion depressed at each level:

- Less than 5 hours: This group has the **highest proportion** of students experiencing depression. The turquoise segment (representing depressed students) dominates the bar, indicating that more than half of students sleeping under 5 hours report depression.

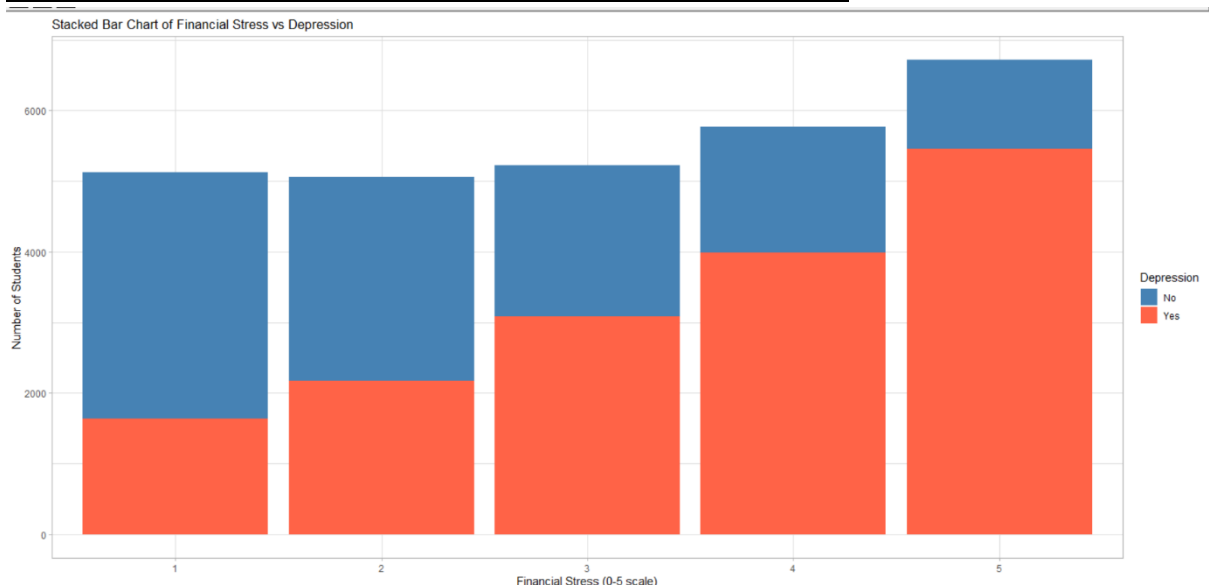
☐ 5–6 hours: Still shows a high proportion of depression, though slightly better than the <5 hours group. Around half the students in this sleep bracket are depressed, pointing to elevated mental health concerns for those getting modestly short sleep.

☐ 7–8 hours: This group demonstrates a **lower proportion** of depression compared to shorter sleep durations. The red segment (non-depressed) is larger, suggesting that students who sleep a typical healthy duration are less likely to be depressed.

☐ More than 8 hours: Similar to the 7–8 hours category, with a larger proportion of students not suffering from depression. This supports the general idea that adequate or slightly longer sleep correlates with better mental health outcomes.

Comment: There is a clear trend across groups indicating that **shorter sleep duration is associated with higher depression levels**, while students with **7+ hours of sleep are less likely to be depressed**. This group-wise insight strongly supports the hypothesis that adequate sleep may act as a protective factor against depression in students.

5. STACKBAR PLOT OF FINANCIAL STRESS VS DEPRESSION



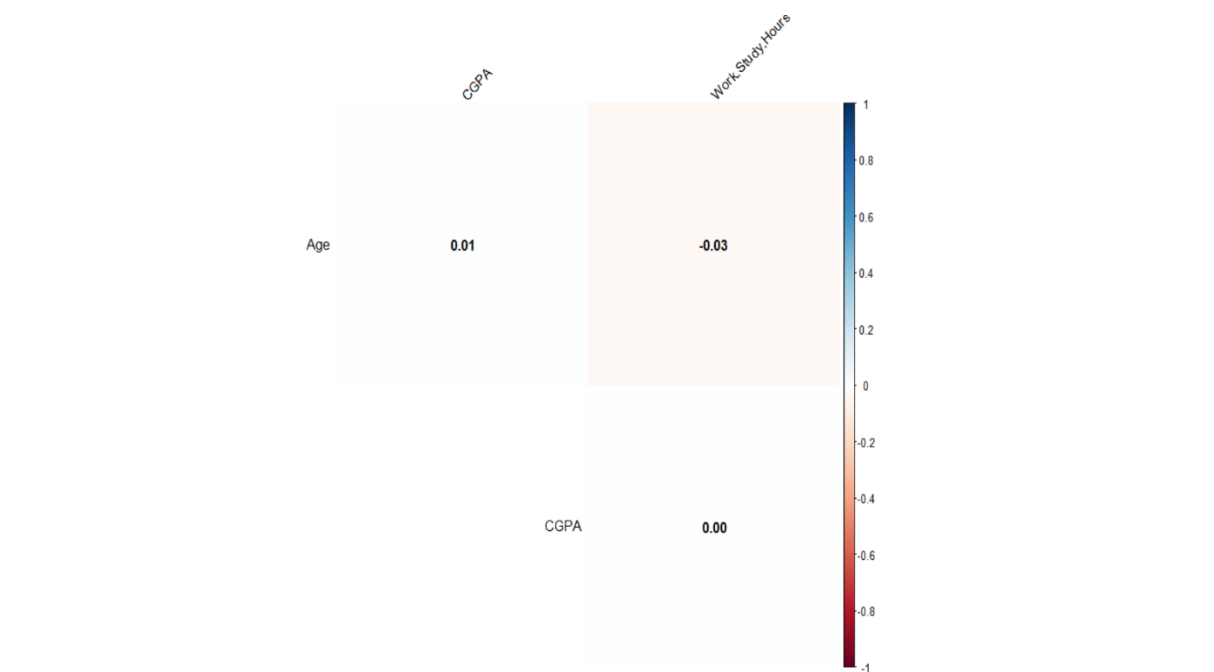
INTERPRETATION OF THE STACKPLOT

The stacked bar chart illustrates the distribution of depression status across different levels of financial stress, measured on a 0–5 scale. Each bar represents a level of financial stress, divided into segments showing students with depression (“Yes”, red) and without depression (“No”, blue).

From the plot, we observe a clear upward trend:

- At **lower levels of financial stress (1 and 2)**, the majority of students are **not depressed**, indicated by the dominant blue segments.
- As financial stress increases to levels 3 and 4, the proportion of students experiencing depression rises sharply, and by level 4, the depressed group nearly equals or surpasses the non-depressed group.
- At the **highest financial stress level (5)**, the situation reverses — the majority of students report being depressed, with the red segment dominating the bar.

6.CORRELATION HEATMAP



INTERPRETATION OF CORRELATION HEATMAP

This correlation heatmap illustrates the pairwise relationships among the numeric variables in our dataset, specifically Age, CGPA, and Work/Study Hours.

- The correlation coefficients are displayed inside the cells, ranging from **-1 to +1**, where:
 - **+1** indicates a perfect positive relationship,
 - **-1** indicates a perfect negative relationship,
 - and **0** indicates no linear relationship.
- **Age vs CGPA:**
Shows a very weak positive correlation (0.01), effectively implying no meaningful linear relationship between a student's age and their CGPA.
- **Age vs Work/Study Hours:**
Displays a very slight negative correlation (-0.03), suggesting that as age increases, work/study hours might decrease marginally, but the relationship is practically negligible.
- **CGPA vs Work/Study Hours:**
Correlation is almost zero (0.00), indicating no linear relationship between CGPA and hours spent studying or working.

Comment: This heatmap indicates that **none of these numeric variables are strongly linearly correlated with each other**. However we shall check for multicollinearity in the later stages.

We conclude our Visual data analysis here and proceed to **Inferential analysis**.

INFERENTIAL ANALYSIS

CHI-SQUARE TESTS AS MEASURES OF ASSOCIATION BETWEEN DEPRESSION STATUS AND INDEPENDENT CATEGORICAL VARIABLES

To examine the association between categorical variables in this study—such as depression status and factors like financial stress, sleep duration, gender, academic pressure, and suicidal thoughts—a series of chi-square tests of independence were conducted. It compares the **observed frequencies** in each category of a contingency table with the **frequencies expected** if there were no association between the variables.

In this analysis, it helps assess whether factors like financial stress levels or sleep duration categories are significantly associated with the prevalence of depression among students. **A significant chi-square result would indicate that the distribution of depression differs meaningfully across the categories of the explanatory variable.**

1. CHI SQ TEST BETWEEN GENDER AND DEPRESSION

We first make a 2x2 contingency table for gender and depression

	0	1	Sum
Female	5132	7220	12352
Male	6431	9115	15546
Sum	11563	16335	27898

After conducting the test, we get the following result:

Pearson's Chi-squared test with Yates' continuity correction

```
data: table_gender_depression
```

```
X-squared = 0.084992, df = 1, p-value = 0.7706
```

INTERPRETATION:

Hypothesis testing

- **H0 (Null Hypothesis):** Gender and Depression are **independent**.
- **H1 (Alternative Hypothesis):** Gender and Depression are **associated** (dependent).

At significance level $\alpha = 0.05$:

- Since **p-value = 0.7706 > 0.05**,
- **we fail to reject the null hypothesis.**

There is **no statistically significant association between gender and depression levels among the students in this sample ($\chi^2 = 0.085$, $p = 0.771$)**. This suggests that the prevalence of depression does not differ meaningfully between male and female students.

2. CHI SQ TEST BETWEEN ACADEMIC PRESSURE AND DEPRESSION

We first construct a 6x2 contingency table for academic pressure and depression

	0	1	Sum
0	5	4	9
1	3868	932	4800
2	2611	1566	4177
3	2973	4488	7461
4	1230	3925	5155
5	876	5420	6296
Sum	11563	16335	27898

After conducting the test we get the following result:

Pearson's Chi-squared test

```
data: table_academic_depression  
X-squared = 6425.2, df = 5, p-value < 2.2e-16
```

INTERPRETATION

Hypothesis Testing:

- **H0 (Null Hypothesis):** Academic pressure and depression are **independent** (no association).
- **H1 (Alternative Hypothesis):** Academic pressure and depression are **associated**.

Using $\alpha = 0.05$:

- Since **p-value < 0.00001 (< 0.05)**, we **reject H0**

There is a **highly significant association between academic pressure and depression among students ($\chi^2 = 6425.2$, $df = 5$, $p < 0.0001$)**. This suggests that levels of academic pressure are strongly related to the prevalence of depression.

3. CHI SQ TEST BETWEEN STUDY SATISFACTION AND DEPRESSION

We first construct a 6x2 contingency table for study satisfaction and depression

	0	1	Sum
0	4	6	10
1	1593	3856	5449
2	2070	3768	5838
3	2467	3353	5820
4	3095	3264	6359
5	2334	2088	4422
Sum	11563	16335	27898

After conducting the test, we get the following result:

```
Pearson's Chi-squared test  
  
data: table_study_depression  
X-squared = 794.14, df = 5, p-value < 2.2e-16
```

INTERPRETATION

HYPOTHESIS TESTING

- **H0 (Null Hypothesis):** Study satisfaction and depression are **independent** (no association).
- **H1 (Alternative Hypothesis):** Study satisfaction and depression are **associated**.

At $\alpha = 0.05$:

- Since **p-value < 0.00001 (< 0.05)**, we reject H0.

There is a **statistically significant association between study satisfaction and depression among students ($\chi^2 = 794.14$, $df = 5$, $p < 0.0001$)**.

This means that how satisfied students are with their studies is strongly linked to whether they experience depression.

4. CHI SQ TEST BETWEEN SLEEP DURATION AND DEPRESSION

We first construct a 5x2 contingency table for sleep duration and depression

	0	1	Sum
5-6 hours	2665	3516	6181
7-8 hours	2975	4371	7346
Less than 5 hours	2948	5361	8309
More than 8 hours	2966	3078	6044
Others	9	9	18
Sum	11563	16335	27898

After conducting the test, we get the result as follows:

Pearson's Chi-squared test

```
data: table_sleep_depression
X-squared = 277.13, df = 4, p-value < 2.2e-16
```

INTERPRETATION

HYPOTHESIS TESTING

- **H0 (Null Hypothesis):** Sleep duration and depression are **independent** (no association).
- **H1 (Alternative Hypothesis):** Sleep duration and depression are **associated**.

At $\alpha = 0.05$:

- Since **p-value < 0.00001 (< 0.05)**, we **reject H0**.

There is a **statistically significant association between sleep duration and depression among students ($\chi^2 = 277.13$, $df = 4$, $p < 0.0001$)**. This indicates that the amount of sleep students get is strongly related to whether they experience depression.

5. CHI SQ TEST BETWEEN DIETARY HABITS AND DEPRESSION

We construct a 4x2 contingency table for diet against depression:

	0	1	Sum
Healthy	4177	3472	7649
Moderate	4363	5558	9921
Others	4	8	12
Unhealthy	3019	7297	10316
Sum	11563	16335	27898

After conducting the test we get the following result:

Pearson's Chi-squared test

```
data: table_diet_depression
X-squared = 1203.3, df = 3, p-value < 2.2e-16
```

INTERPRETATION

HYPOTHESIS TESTING

- **H0 (Null Hypothesis):** Dietary habits and depression are **independent** (no relationship).
- **H1 (Alternative Hypothesis):** Dietary habits and depression are **associated**.

At significance level $\alpha = 0.05$:

- Since **p-value < 0.0001 (< 0.05)**, we **reject H0**.

There is a **statistically significant association between dietary habits and depression among students** ($\chi^2 = 1203.3$, $df = 3$, $p < 0.0001$). This means students' eating habits are strongly linked with their likelihood of experiencing depression.

6. CHI SQ TEST BETWEEN SUICIDAL THOUGHTS AND DEPRESSION

We construct a 2x2 contingency table for suicidal thoughts against depression

	0	1	Sum
No	7864	2378	10242
Yes	3699	13957	17656
Sum	11563	16335	27898

After conducting the test we get the following result:

Pearson's Chi-squared test with Yates' continuity correction

```
data: table_suicide_depression
X-squared = 8323.4, df = 1, p-value < 2.2e-16
```

INTERPRETATION

HYPOTHESIS TESTING

- **H0 (Null Hypothesis):** Having ever had suicidal thoughts and depression are **independent** (not associated).
- **H1 (Alternative Hypothesis):** They are **associated**.

At $\alpha = 0.05$:

- The **p-value** < 0.0001 , which is much smaller than 0.05.
- Therefore, we **reject H0**.

There is a very strong statistical association between having ever had suicidal thoughts and depression ($\chi^2 = 8323.4$, $df = 1$, $p < 0.0001$). Students who have experienced suicidal thoughts are significantly more likely to also show signs of depression.

7. CHI SQ TEST BETWEEN FINANCIAL STRESS AND DEPRESSION

We construct a 5x2 contingency table for financial stress and depression

	0	1	Sum
1	3489	1632	5121
2	2886	2175	5061
3	2146	3080	5226
4	1785	3990	5775
5	1257	5458	6715
Sum	11563	16335	27898

After conducting the test, we get the following results:

```
Pearson's Chi-squared test

data:  table_financial_depression
X-squared = 3702.4, df = 4, p-value < 2.2e-16
```

INTERPRETATION

HYPOTHESIS TESTING

- **H0 (Null Hypothesis):** Financial stress and depression are **independent** (there is no association).
- **H1 (Alternative Hypothesis):** Financial stress and depression are **associated** (they are not independent).

At a **significance level of $\alpha = 0.05$:**

- If **p-value < 0.05**, reject H0.

There is a statistically significant association between financial stress and depression among students ($\chi^2 = 3702.4$, $df = 4$, $p < 0.0001$). This means that students reporting higher levels of financial stress are considerably more likely to experience symptoms of depression.

8. CHI SQ TEST BETWEEN FAMILY HISTORY AND DEPRESSION

We construct a 2x2 contingency table for family history and depression

	0	1	Sum
No	6335	8062	14397
Yes	5228	8273	13501
Sum	11563	16335	27898

After conducting the test, we get the following result:

```
Pearson's Chi-squared test with Yates' continuity correction

data: table_family_depression
X-squared = 79.794, df = 1, p-value < 2.2e-16
```

INTERPRETATION

HYPOTHESIS TESTING

- **H0 (Null Hypothesis):** Family support and depression are **independent** (no association).
- **H1 (Alternative Hypothesis):** Family support and depression are **associated** (not independent).

At a **significance level of $\alpha = 0.05$:**

- If **p-value < 0.05**, reject H0.

There is a statistically significant association between family history and depression among students ($\chi^2 = 79.8$, $df = 1$, $p < 0.0001$).

This suggests that students who have less family history of mental illness are significantly less likely to experience symptoms of depression.

SUMMARY TABLE

Factor vs Depression	χ^2 Statistic	df	p-value	Association
Gender	0.085	1	0.7706	Not significant
Academic Pressure	6425.2	5	< 0.0001	Significant
Study Satisfaction	794.1	5	< 0.0001	Significant
Sleep Duration	277.1	4	< 0.0001	Significant
Dietary Habits	1203.3	3	< 0.0001	Significant
Suicidal Thoughts	8323.4	1	< 0.0001	Highly significant
Financial Stress	3702.4	4	< 0.0001	Significant
Family Support	79.8	1	< 0.0001	Significant

The chi-square tests revealed that all factors examined, except gender, showed statistically significant associations with depression among students ($p < 0.0001$ in most cases), indicating that variables such as academic pressure, study satisfaction, sleep duration, dietary habits, suicidal thoughts, financial stress, and family support are closely linked to depression prevalence.

TEST FOR CGPA VS DEPRESSION:

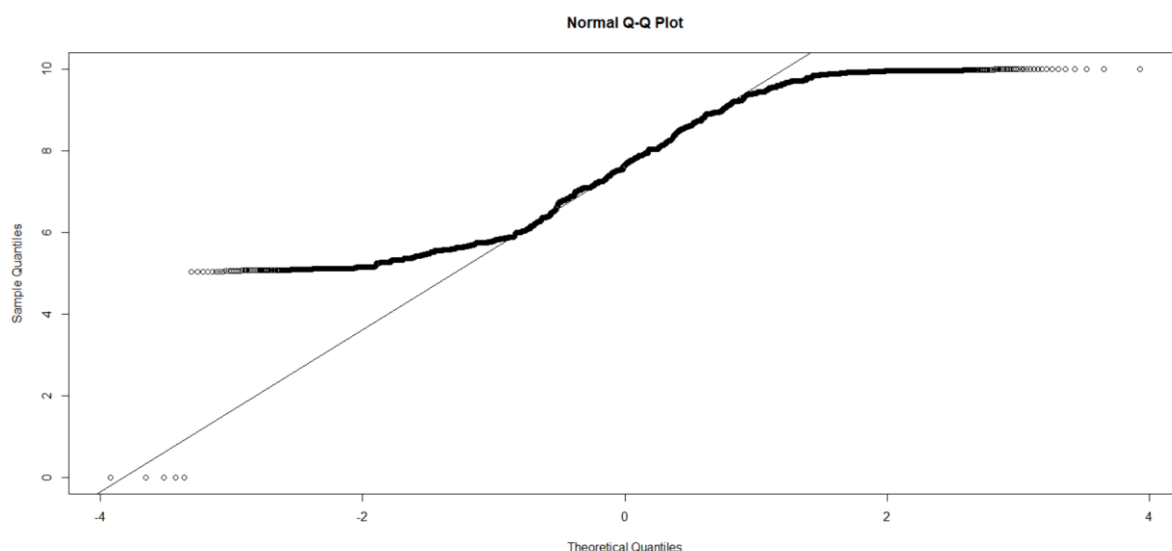
To examine whether academic performance, as measured by CGPA, is associated with depression among students, an independent samples test is conducted. Given that CGPA is a continuous variable and depression is a

categorical variable with two levels (not depressed and depressed), this analysis compares the average CGPA scores across the two depression groups to determine if a significant difference exists.

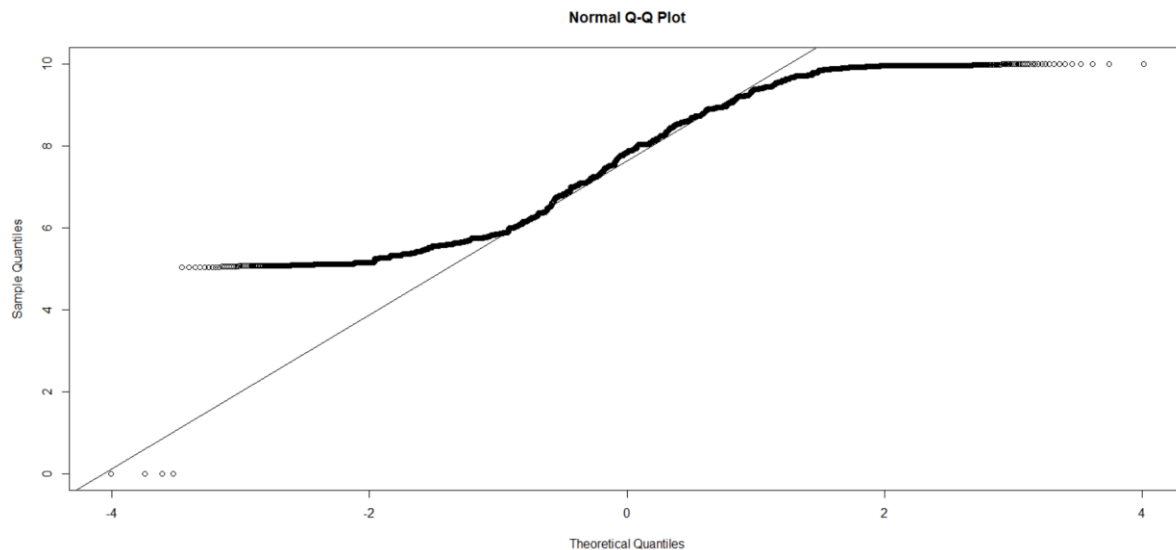
To examine whether academic performance, as measured by CGPA, differs between students with and without depression, an **independent samples t-test** can be performed. This test compares the mean CGPA scores of the two groups to determine if any statistically significant difference exists, under the assumption of approximately normal distribution of CGPA within groups.

However if normality does not hold, Wilcoxon rank-sum test (Mann-Whitney U test) has to be conducted to assess whether academic performance varies between students with and without depression. This non-parametric test compares the distribution of CGPA scores across the two depression groups to evaluate the presence of any significant differences.

Using **Q-Q plot** to check for normality,



QQ plot for CGPA for students not depressed



QQ plot for CGPA for students depressed

From the plots we can say:

On the left side (lower quantiles), our data are below the line, suggesting heavier left tail (or outliers). In the middle region, the points rise more slowly than expected under normality (they are above the line for quite a stretch), indicating data might be skewed or have a plateau effect. On the right side, the points again deviate above the line. Overall, this indicates **our data does not follow a normal distribution**.

So we proceed to the **non parametric Wilcoxon rank sum test**.

We get the following result:

Wilcoxon rank sum test with continuity correction

data: CGPA by Depression

W = 92023460, p-value = 0.0002643

alternative hypothesis: true location shift is not equal to 0

INTERPRETATION

HYPOTHESIS

- **H0 (Null Hypothesis):** The distributions of CGPA are the **same across depression groups** (no difference in academic performance).
- **H1 (Alternative Hypothesis):** The distributions of CGPA **differ between depression groups** (i.e., median CGPA differs).

Since **p-value = 0.0002643 < 0.05**, we **reject the null hypothesis H0**. This indicates that there is a **statistically significant difference in CGPA between students with and without depression**.

TEST FOR AGE VS DEPRESSION

The primary objective of this analysis is to investigate whether the proportion of students experiencing depression differs across various age groups. By dividing age into **three increasing categories**, this test seeks to determine if age is associated with depression prevalence among students. Understanding this relationship can provide insights into whether certain age groups are more vulnerable to depression, thereby helping to identify specific cohorts that may benefit from targeted mental health interventions.

The 3 age groups we have considered are : **<=20, 21-23 and >23**

The table showing the number of depressed and non depressed students in each category is given below:

```
> print(table_age_depression)
```

	0	1
≤20	1488	3895
21-23	1610	2921
>23	8465	9519

After conducting the test we get the proportion of depressed students in each group as:

```
3-sample test for equality of proportions without continuity correction

data:  x out of n
X-squared = 722.26, df = 2, p-value < 2.2e-16
alternative hypothesis: two.sided
sample estimates:
  prop 1    prop 2    prop 3 
0.7235742 0.6446701 0.5293038
```

INTERPRETATION

HYPOTHESIS TESTING

- **(H₀):**
The proportion of students experiencing depression is equal across all three age groups (≤ 20 , 21–23, and > 23 years).
- **(H₁):**
The proportion of students experiencing depression differs in at least one age group, indicating an association between age and depression prevalence.

The test for equality of binomial proportions revealed a **statistically significant difference in depression prevalence across the three age groups hence rejecting H₀.**

This suggests that the **proportion of students experiencing depression is not consistent across age categories.**

Specifically, the highest proportion of depression was observed among students aged **≤ 20 years (72.4%)**, followed by those aged **21–23 years (64.5%)**, with the lowest prevalence noted in students aged **> 23 years (52.9%)**.

These findings indicate that **younger students are more susceptible to depression**, highlighting the importance of early mental health interventions targeted at this demographic.

We conclude our Inferential study here and proceed to analysis of **Logistic model.**

LOGISTIC REGRESSION

To further explore the factors associated with depression among students, a **binary logistic regression analysis** is performed. This method allows for examining the influence of multiple predictor variables simultaneously on the likelihood of experiencing depression, **treating depression status as a binary outcome (depressed vs not depressed)**. By estimating odds ratios, logistic regression helps identify which factors significantly increase or decrease the probability of depression, thereby providing deeper insights beyond simple bivariate comparisons.

Let us denote the response variable Depression by Y . Here Y is a binary variable denoting depression status of a student under study.

We assume $Y_i \sim \text{Bernoulli}(p_i)$, $i=1(1)27898$, where p_i denotes the **probability of i th student being depressed**

The rest of the 11 variables which are to be used as predictors in our model are defined as follows:

- X_1 : Gender
- X_2 :Age
- X_3 : Academic Pressure ($X_{31}, X_{32}, X_{33}, X_{34}, X_{35}$ denote the levels pressure)
- X_4 : CGPA
- X_5 : Study.Satisfaction ($X_{51}, X_{52}, X_{53}, X_{54}, X_{55}$ denote the levels of satisfaction)
- X_6 : Sleep.Duration ($X_{61}, X_{62}, X_{63}, X_{64}$ denote the levels of sleep)
- X_7 : Dietary.Habits (X_{71}, X_{72}, X_{73} denote the levels of diet)
- X_8 : Work.Study.Hours
- X_9 : Financial.Stress ($X_{92}, X_{93}, X_{94}, X_{95}$ denote the levels of stress)
- X_{10} : Family.History.of.Mental.Illness
- X_{11} : Have.you.ever.had.suicidal.thoughts

MODEL:

$$\begin{aligned} \log(p / (1 - p)) = & \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \gamma_1 X_{31} + \gamma_2 X_{32} + \gamma_3 X_{33} + \gamma_4 X_{34} + \gamma_5 X_{35} \\ & + \beta_3 X_4 + \delta_1 X_{51} + \delta_2 X_{52} + \delta_3 X_{53} + \delta_4 X_{54} + \delta_5 X_{55} + \theta_1 X_{61} + \theta_2 X_{62} + \theta_3 X_{63} + \\ & \theta_4 X_{64} + \lambda_1 X_{71} + \lambda_2 X_{72} + \lambda_3 X_{73} + \beta_5 X_8 + \eta_1 X_{92} + \eta_2 X_{93} + \eta_3 X_{94} + \eta_4 X_{95} + \\ & \beta_6 X_{10} + \beta_7 X_{11} + e \end{aligned}$$

where

- β : Coefficients for main predictors such as Gender (X_1), Age (X_2), CGPA (X_4), Work-Study Hours (X_8), Family History of Mental Illness (X_{10}), and Suicidal Thoughts (X_{11}).
- γ : Coefficients for levels of Academic Pressure (X_{31} to X_{35}).
- δ : Coefficients for levels of Study Satisfaction (X_{51} to X_{55}).
- θ : Coefficients for levels of Sleep Duration (X_{61} to X_{64}).
- λ : Coefficients for levels of Dietary Habits (X_{71} to X_{73}).
- η : Coefficients for levels of Financial Stress (X_{92} to X_{95}).
- e : random error

Now we shall proceed to fit the model using R.

The model summary is as follows:

```
Call:
glm(formula = Depression ~ Gender + CGPA + Academic.Pressure +
    Sleep.Duration + Study.Satisfaction + Financial.Stress +
    Family.History.of.Mental.Illness + Have.you.ever.had.suicidal.thoughts +
    Age + Dietary.Habits + Work.Study.Hours, family = binomial,
    data = datal)

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)      -2.378993    1.019425  -2.334  0.0196
GenderMale         -0.009784    0.036770  -0.266  0.7902
CGPA                0.063318    0.012430   5.094 3.51e-07
Academic.Pressure1  -0.623218    1.294881  -0.481  0.6303
Academic.Pressure2   0.384814    1.294735   0.297  0.7663
Academic.Pressure3   1.271402    1.294413   0.982  0.3260
Academic.Pressure4   2.058874    1.294514   1.590  0.1117
Academic.Pressure5   2.748564    1.294697   2.123  0.0338
Sleep.Duration7-8 hours  0.018663    0.051831   0.360  0.7188
Sleep.DurationLess than 5 hours  0.363494    0.051234   7.095 1.29e-12
Sleep.DurationMore than 8 hours -0.255399    0.054341  -4.700 2.60e-06
Sleep.DurationOthers -0.250188    0.656830  -0.381  0.7033
Study.Satisfaction1  0.279177    1.458717   0.191  0.8482
Study.Satisfaction2 -0.151292    1.458293  -0.104  0.9174
Study.Satisfaction3 -0.208036    1.458158  -0.143  0.8866
Study.Satisfaction4 -0.557504    1.458469  -0.382  0.7023
Study.Satisfaction5 -0.762569    1.458578  -0.523  0.6011
Financial.Stress2     0.431737    0.057969   7.448 9.50e-14
Financial.Stress3     1.084704    0.057818  18.761 < 2e-16
Financial.Stress4     1.550160    0.057742  26.846 < 2e-16
Financial.Stress5     2.213112    0.060033  36.865 < 2e-16
Family.History.of.Mental.IllnessYes 0.244919    0.036456   6.718 1.84e-11
Have.you.ever.had.suicidal.thoughtsYes 2.499873    0.039059  64.002 < 2e-16
Age                  -0.110069    0.003837 -28.683 < 2e-16
Dietary.HabitsModerate 0.472515    0.045144  10.467 < 2e-16
Dietary.HabitsOthers  1.020827    0.771166   1.324  0.1856
Dietary.HabitsUnhealthy 1.091944    0.046376  23.545 < 2e-16
Work.Study.Hours      0.117482    0.004978  23.601 < 2e-16

Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 37855  on 27897  degrees of freedom
Residual deviance: 19439  on 27870  degrees of freedom
AIC: 19495

Number of Fisher Scoring iterations: 6
```

So by Fisher scoring, we get the estimates of model parameters as:

$$\begin{aligned} \beta_0 &= -2.378 & \beta_1 &= -0.098 & \beta_2 &= -0.110 & \beta_3 &= 0.063 & \beta_5 &= 0.117 & \beta_6 &= 0.244 \\ \beta_7 &= 2.499 & \gamma_1 &= -0.623 & \gamma_2 &= 0.384 & \gamma_3 &= 1.271 & \gamma_4 &= 2.059 & \gamma_5 &= \\ & 2.749 & \delta_1 &= 0.279 & \delta_2 &= -0.151 & \delta_3 &= -0.208 & \delta_4 &= -0.055 & \delta_5 &= -0.762 \\ \theta_1 &= 0.018 & \theta_2 &= 0.363 & \theta_3 &= -0.255 & \theta_4 &= -0.250 & \lambda_1 &= 0.473 & \lambda_2 &= \\ & 1.021 & \lambda_3 &= 1.091 & \eta_1 &= 0.431 & \eta_2 &= 1.084 & \eta_3 &= 1.550 & \eta_4 &= 2.213 \end{aligned}$$

The fitted model is:

$$\log\left(\frac{p}{p-1}\right) = -2.378 - 0.009x_1 - 0.110x_2 - 0.623x_{31} + 0.384x_{32} + 1.271x_{33} + 2.059x_{34} + 2.749x_{35} + 0.063x_4 + 0.279x_{51} - 0.151x_{52} - 0.208x_{53} - 0.055x_{54} - 0.762x_{55} + 0.018x_{61} + 0.363x_{62} - 0.255x_{63} - 0.250x_{64} + 0.473x_{71} + 1.021x_{72} + 1.091x_{73} + 0.117x_8 + 0.431x_{92} + 1.084x_{93} + 1.550x_{94} + 2.213x_{95} + 0.244x_{10} + 2.499x_{11}$$

INTERPRETATION OF MODEL PARAMETERS

$\beta_0 = -2.378$ means that when all the predictors are at their reference categories (or zero), the log odds of a student being depressed is -2.378 times the log odds of the student not being depressed.

$\beta_1 = -0.098$ (Gender) Holding all other variables constant, being male decreases the log odds of being depressed by **0.098 times** compared to being female.

$\beta_2 = -0.110$ (Age) For each additional year of age, the log odds of being depressed decreases by **0.110 times**, holding other predictors constant.

$\beta_3 = 0.063$ (CGPA) For each one-unit increase in CGPA, the log odds of being depressed increases by **0.063 times**, assuming all other variables remain unchanged.

$\beta_5 = 0.117$ (Work-Study Hours) For each additional unit of work-study hours, the log odds of being depressed increases by **0.117 times**, controlling for other factors.

$\beta_6 = 0.244$ (Family History of Mental Illness) Having a family history of mental illness increases the log odds of being depressed by **0.244 times** compared to not having such a history.

$\hat{\beta}_7 = 2.499$ (**Ever had Suicidal Thoughts**) Having ever had suicidal thoughts increases the log odds of being depressed by **2.499 times** compared to those who have not.

$\hat{\gamma}_1 = -0.623$ Compared to students experiencing the lowest level of academic pressure (reference group), students at the first higher level of academic pressure have log odds of being depressed that are **0.623 times lower**.

$\hat{\gamma}_2 = 0.384$ Compared to the reference group, students at the second level of academic pressure have log odds of being depressed that are **0.384 times higher**.

$\hat{\gamma}_3 = 1.271$ Compared to the reference group, students at the third level of academic pressure have log odds of being depressed that are **1.271 times higher**.

$\hat{\gamma}_4 = 2.059$ Compared to the reference group, students at the fourth level of academic pressure have log odds of being depressed that are **2.059 times higher**.

$\hat{\gamma}_5 = 2.749$ Compared to the reference group, students experiencing the highest level of academic pressure have log odds of being depressed that are **2.749 times higher**.

$\hat{\delta}_1 = 0.279$ Compared to students who are **least satisfied with their studies** (reference group, level 0), students at the first higher level of study satisfaction have log odds of being depressed that are **0.279 times higher**.

$\hat{\delta}_2 = -0.151$ Compared to the least satisfied students, students at the second level of study satisfaction have log odds of being depressed that are **0.151 times lower**.

$\hat{\delta}_3 = -0.208$ Compared to the least satisfied students, students at the third level of study satisfaction have log odds of being depressed that are **0.208 times lower**.

$\hat{\delta}_4 = -0.055$ Compared to the least satisfied students, students at the fourth level of study satisfaction have log odds of being depressed that are **0.055 times lower**.

$\delta_5 = -0.762$ Compared to the least satisfied students, students at the highest level of study satisfaction have log odds of being depressed that are **0.762 times lower**.

$\hat{\theta}_1 = 0.018$ Compared to students who sleep **5–6 hours** (reference group), students who sleep **7–8 hours** have log odds of being depressed that are **0.018 times higher**, essentially indicating **no meaningful difference**.

$\hat{\theta}_2 = 0.363$ Compared to the reference group, students who sleep **less than 5 hours** have log odds of being depressed that are **0.363 times higher**, suggesting **increased odds of depression**.

$\hat{\theta}_3 = -0.255$ Compared to the reference group, students who sleep **more than 8 hours** have log odds of being depressed that are **0.255 times lower**, indicating a **mild reduction in odds**.

$\hat{\theta}_4 = -0.250$ Compared to the reference group, students in the **‘Other’ sleep category** have log odds of being depressed that are **0.250 times lower**.

$\hat{\lambda}_1 = 0.473$ (**Moderate Diet**) Compared to students who follow a **healthy diet** (reference group), students with a **moderate diet** have log odds of being depressed that are **0.473 times higher**, indicating moderately increased odds of depression.

$\hat{\lambda}_3 = 1.021$ (**Unhealthy Diet**) Compared to the reference group, students with an **unhealthy diet** have log odds of being depressed that are **1.021 times higher**, suggesting substantially increased odds of depression.

$\hat{\lambda}_2 = 1.091$ (**Others**) Compared to the reference group, students in the **‘Other’ dietary category** have log odds of being depressed that are **1.091 times higher**, also indicating higher odds of depression.

$\hat{\eta}_2 = 0.431$ (**Level 2 Financial Stress**) Compared to students experiencing the **lowest level of financial stress** (reference group), students at **level 2 financial stress** have log odds of being depressed that are **0.431 times higher**, indicating moderately increased odds.

$\hat{\eta}_3 = 1.084$ (**Level 3 Financial Stress**) Compared to the reference group, students at **level 3 financial stress** have log odds of being depressed that are **1.084 times higher**, showing a strong increase in odds of depression.

$\hat{\eta}_4 = 1.550$ (**Level 4 Financial Stress**) Compared to the reference group, students at **level 4 financial stress** have log odds of being depressed that are **1.550 times higher**, indicating substantially elevated odds.

$\hat{\eta}_5 = 2.213$ (**Level 5 Financial Stress**) Compared to the reference group, students at **level 5 (highest) financial stress** have log odds of being depressed that are **2.213 times higher**, reflecting the strongest increase in odds.

The model highlights that **younger age, higher CGPA, extreme financial stress, family history of mental illness, having suicidal thoughts, unhealthy or moderate diet, insufficient sleep (<5 hrs), and very high academic pressure** are all statistically significant factors associated with increased odds of depression.

MULTICOLLINEARITY

Multicollinearity happens when two or more predictor variables within a regression model are correlated highly, which implies that they hold redundant or overlapping information.

Multicollinearity makes it challenging to decide the effect of each predictor individually on the response variable.

We use the **vif()** function from the “car” package in R to obtain the **Variance Inflation Factor (VIF)**. As a rule of thumb, **VIF > 5** means **multicollinearity exists**. The **Variance Inflation Factor (VIF)** indicates to what extent the variance of a regression coefficient is inflated because of multicollinearity. That is, it informs us if a predictor is highly correlated with other predictors in the model.

If VIF value of a variable is high we will have to remove that variable and construct a new model.

We have obtained the following values

```
> vif(model)
```

	GVIF	Df	GVIF^(1/(2*Df))
Gender	1.010845	1	1.005408
CGPA	1.015143	1	1.007543
Academic.Pressure	2.428174	5	1.092768
Sleep.Duration	1.017081	4	1.002119
Study.Satisfaction	2.268779	5	1.085374
Financial.Stress	1.056024	4	1.006837
Family.History.of.Mental.Illness	1.003195	1	1.001596
Have.you.ever.had.suicidal.thoughts	1.086248	1	1.042232
Age	1.035716	1	1.017701
Dietary.Habits	1.030320	3	1.004991
Work.Study.Hours	1.017846	1	1.008884

The VIF values are well **below the commonly used thresholds of 5 or 10**, indicating that **multicollinearity is not a concern in this model**. Thus, the estimated coefficients are expected to be stable and reliable, with **minimal inflation of standard errors due to correlations among predictors**. Hence we can **continue with our original model**.

GOODNESS OF FIT OF THE MODEL

After fitting the logistic model we get that

Null deviance : 37855 on 27897 degrees of freedom

Residual deviance: 19439 on 27870 degrees of freedom

Interpretation:

1. **Null Deviance**: The null deviance reflects the goodness of fit of a model that **only includes the intercept (the null model)**, meaning it has **no predictors**. It shows the **difference in deviance between the null model and the model being assessed**. A lower null deviance suggests that the model accounts for a greater proportion of the variance in the response variable compared to the null model.

2. **Residual Deviance**: The residual deviance indicates the goodness of fit of the **model after adding predictors**. It represents the difference in deviance between the evaluated model and the saturated model (which perfectly fits the data by having a parameter for each observation). **A smaller residual deviance signifies a better fit of the model to the data**.

The large drop from 37855 to 19439 shows that including the predictors (Gender, Age, Academic Pressure, CGPA, etc.) **greatly improves the model's ability to explain or predict depression status among students**.

Comment: Our fitted model explains a substantial portion of the variation in depression status compared to a model without predictors, indicating that the included variables have significant explanatory power. Hence the model has a **good fit**.

QUALITY EVALUATION OF THE MODEL

We evaluate the quality of the logistic regression model by forming a **Confusion Matrix**.

Definition of a Confusion Matrix: A confusion matrix is a performance evaluation tool used in classification models to assess how well the model distinguishes between different classes. It provides a 2x2 tabular summary of predictions by comparing actual and predicted values, offering insights into the types of errors the model makes. By breaking down predictions into categories such as **true positive rate (TPR)**, **false positive rate (FPR)**, **the confusion matrix helps in calculating key evaluation metrics like accuracy, precision etc.** This detailed breakdown enables a deeper understanding of model performance beyond a single accuracy measure.

Choice of threshold: Then we need to choose an **optimum threshold p^*** . The value of p^* is so chosen that for $p_i = p^*$ the True Positive Rate (TPR) is high and the False Positive Rate (FPR) is low. We take p^* for which $TPR^*(1-FPR)$ is maximum. Where, $TPR = P[\text{predicted response} = 1 \mid \text{observed response} = 1]$
 $FPR = P[\text{predicted response} = 1 \mid \text{observed response} = 0]$ Here we get our optimal threshold or cutoff point as $p^* = 0.5087012 = 0.5$ (approximately)

By using the optimum threshold value of 0.5, we form a confusion matrix. We use the **conf_matrix function in R**.

The confusion matrix is:

```
> print(conf_matrix)
      Actual
Predicted  0    1
      0  9158 1857
      1  2405 14478
> |
```

From this matrix we obtain TPR and FPR values as follows:

```
> # TPR = Sensitivity = TP / (TP + FN)
> TPR <- TP / (TP + FN)
>
> # FPR = FP / (FP + TN)
> FPR <- FP / (FP + TN)
>
> # Print results
> cat("TPR:", TPR, "\n")
TPR: 0.8863177
> cat("FPR:", FPR, "\n")
FPR: 0.207991
```

True Positive Rate (TPR) = 0.8863

Also known as **sensitivity** or **recall**, this means:

88.6% of all students who are actually depressed were correctly identified by the model as depressed.

This is quite high, indicating the model does a good job catching most of the depressed students.

False Positive Rate (FPR) = 0.2080

This means:

20.8% of students who are *not depressed* were incorrectly predicted by the model as depressed.

In other words, there is about a 21% chance of mistakenly flagging a non-depressed student as depressed.

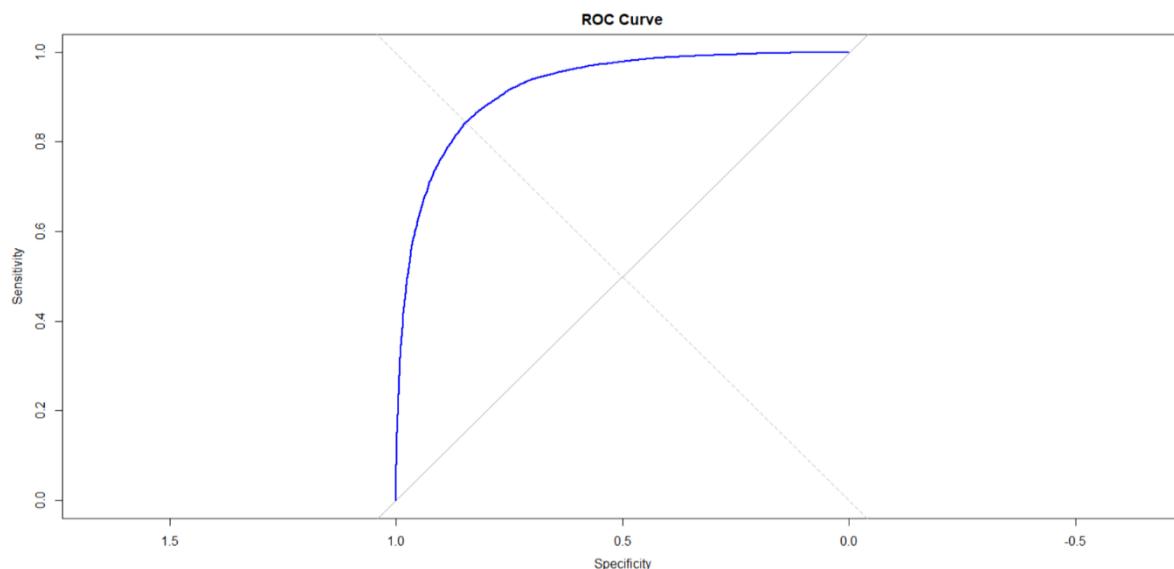
Comment: The model is **sensitive and good at detecting depression** given the high TPR value, with very few trade-off of **false positives**, which might be acceptable depending on the application.

RECEIVER OPERATING CHARACTERISTIC (ROC) CURVE

The Receiver Operating Characteristic (ROC) curve is a **graphical method for measuring the performance of a binary classification model**. ROC curve is used to plot True Positive Rate (TPR) vs False Positive Rate (FPR) at different decision thresholds. The ROC curve offers information on how well the model can separate positive and negative classes at different threshold values.

The **TPR is plotted on the y-axis, while the FPR is plotted on the x-axis. An ROC curve for a model that is closer to the top-left of the plot signifies high sensitivity and low false positive rate, which indicates good model performance.**

We obtain the ROC curve in R which is shown below



Comment: From the curve, we can see: The **blue curve stays well above the diagonal (random classifier line)**, indicating **our model performs significantly better than chance**. The **curve bends quickly toward the top-left and hugs the top left corner of the plot which is a sign of a strong classifier**. If the line was closer to the diagonal it would have indicated poor performance. The ROC curve for our logistic regression model demonstrates a **high degree of separation between the depressed and non-depressed students**.

Further **Area under curve (AUC)** obtained from ROC = **0.9219**. Closer the AUC to 1, better the model. This graphical performance measure, along with the high AUC of **0.9219**, confirms that our logistic regression model possesses **excellent discriminative ability** to correctly classify students as depressed or not depressed.

CONCLUSION

This study sought to unravel the complex interplay of demographic, academic, lifestyle, and psychosocial factors contributing to depression among students. **Through meticulous statistical analysis of a comprehensive dataset—including exploratory data visualizations, chi-square tests, non-parametric comparisons, binomial proportion tests, and a multiple logistic regression model—the project identified several critical determinants of depression within the student population.**

The findings underscore that higher levels of academic pressure, severe financial stress, unhealthy dietary habits, reduced sleep duration, a family history of mental illness, and suicidal ideation significantly elevate the likelihood of depression among students. The logistic regression model demonstrated **excellent predictive capacity, evidenced by a high area under the ROC curve (AUC $\approx$ 0.92), along with favorable sensitivity and specificity, highlighting its robustness in classifying depression status.**

FUTURE SCOPE:

- **Incorporation of longitudinal data:**
Future studies could track students over time to examine how depression risk factors evolve.
- **Intervention-based analysis:**
Implementing and evaluating targeted programs (like stress management workshops or peer support initiatives) could help assess their direct impact on reducing depression.
- **Inclusion of qualitative insights:**
Complementing quantitative models with interviews or focus groups might provide deeper understanding of personal and social contexts influencing depression.
- **Expansion to diverse populations:**
Applying the methodology across different universities, regions, or cultural backgrounds would test the generalizability of findings.
- **Use of advanced machine learning models:**
Techniques such as random forests or neural networks could uncover complex, non-linear interactions between predictors and depression risk.
- **Integration of additional health indicators:**
Including measures like physical activity, substance use, or stress biomarkers may provide a more holistic view of student well-being.

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APPENDIX

DATASET SOURCE:

The **STUDENT DEPRESSION dataset** used in this study was obtained from the website **Kaggle.com** under the title **“Student Depression Dataset”**. The original dataset can be accessed at the link below:

<https://www.kaggle.com/datasets/adilshamim8/student-depression-dataset>

USED TOOLS:

- MS EXCEL
- MS WORD
- R 4.5.0
- GOOGLE

CODES

Data cleaning and preprocessing:

```
rm(list=ls())
getwd()
data<-read.csv(file.choose())
anyNA(data)
data1<-na.omit(data)
anyNA(data1)
data1 <- data1[ , -c(1,4, 5, 7,10,13)]
# Convert key variables to factor
data1$Depression <- as.factor(data1$Depression)
data1$Gender <- as.factor(data1$Gender)
data1$Sleep.Duration <- as.factor(data1$Sleep.Duration)
data1$Financial.Stress <- as.factor(data1$Financial.Stress)
data1$Family.History.of.Mental.Illness <- as.factor(data1$Family.History.of.Mental.Illness)
data1$Academic.Pressure <- as.factor(data1$Academic.Pressure)
data1$Study.Satisfaction <- as.factor(data1$Study.Satisfaction)
data1$Have.you.ever.had.suicidal.thoughts<-as.factor(data1$Have.you.ever.had.suicidal.thoughts)
data1$Dietary.Habits<-as.factor(data1$Dietary.Habits)
str(data1)
```

Data visualisation:

```
ggplot(data1, aes(x = factor(Depression), y = CGPA)) +
  geom_boxplot(fill = "tomato", alpha = 0.6) +
  stat_summary(fun = mean, geom = "point", shape = 20, size = 3, color = "darkblue") +
```

```

labs( title = "Boxplot of CGPA by Depression Status with Means", x = "Depression (0 = No, 1 = Yes)", y = "CGPA )
+
  theme_light()
library(ggplot2)
ggplot(data1, aes(x = factor(Academic Pressure), fill = factor(Depression))) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(values = c("steelblue", "tomato"),
                    name = "Depression",
                    labels = c("No", "Yes")) +
  labs( title = "Stacked Proportion Bar Chart of Academic Pressure vs Depression", x = "Academic Pressure (0-5)
y = "Proportion"
) +
  theme_minimal()
library(ggplot2)
ggplot(data1, aes(x = Gender, fill = factor(Depression))) +
  geom_bar() +
  scale_fill_manual(values = c("steelblue", "tomato"),
                    name = "Depression",
                    labels = c("No", "Yes")) +
  labs(
    title = "Stacked Bar Chart of Gender vs Depression",
    x = "Gender",
    y = "Count"
  ) +
  theme_light()
library(ggplot2)
# Install if needed
# install.packages("GGally")
library(GGally)
# Make sure Depression is a factor
data1$Depression <- factor(data1$Depression, labels = c("No", "Yes"))
# Select variables for pair plot
pair_data <- data1[, c("CGPA", "Academic.Pressure", "Financial.Stress",
                      "Study.Satisfaction", "Work.Study.Hours", "Depression")]

# Create the pair plot
ggpairs(pair_data, aes(color = Depression, alpha = 0.5))
library(ggplot2)
ggplot(data1, aes(x = factor(Financial.Stress), fill = factor(Depression))) +
  geom_bar() +
  scale_fill_manual(values = c("steelblue", "tomato"),
                    name = "Depression",
                    labels = c("No", "Yes")) +
  labs(title = "Stacked Bar Chart of Financial Stress vs Depression", x = "Financial Stress (0-5 scale)",
    y = "Number of Students"
  ) +
  theme_light()
library(ggplot2)
# Make sure Depression is a factor
data1$Depression <- as.factor(data1$Depression)
# Stacked bar plot with counts
ggplot(data1, aes(x = Academic.Pressure, fill = Depression)) +

```

```
geom_bar(position = "stack") +
labs(x = "Academic Pressure Level", y = "Number of Students",
fill = "Depression", title = "Depression by Academic Pressure") +
theme_minimal()
```

Testing of hypotheses:

```
table_gender_depression<-table(data1$Gender,data1$Depression)
print(table_gender_depression)
addmargins(table_gender_depression)
chi_result1<-chisq.test(table_gender_depression)
print(chi_result1)
table_academic_depression<-table(data1$Academic.Pressure,data1$Depression)
print(table_academic_depression)
addmargins(table_academic_depression)
chi_result5<-chisq.test(table_academic_depression)
print(chi_result5)
table_study_depression<-table(data1$Study.Satisfaction,data1$Depression)
print(table_study_depression)
addmargins(table_study_depression)
chi_result6<-chisq.test(table_study_depression)
print(chi_result6)
table_sleep_depression<-table(data1$Sleep.Duration,data1$Depression)
print(table_sleep_depression)
addmargins(table_sleep_depression)
chi_result3<-chisq.test(table_sleep_depression)
print(chi_result3)
table_diet_depression<-table(data1$Dietary.Habits,data1$Depression)
print(table_diet_depression)
addmargins(table_diet_depression)
chi_result7<-chisq.test(table_diet_depression)
print(chi_result7)
table_suicide_depression<-table(data1$Have.you.ever.had.suicidal.thoughts,data1$Depression)
print(table_suicide_depression)
addmargins(table_suicide_depression)
chi_result4<-chisq.test(table_suicide_depression)
print(chi_result4)
table_financial_depression<-table(data1$Financial.Stress,data1$Depression)
print(table_financial_depression)
addmargins(table_financial_depression)
chi_result2<-chisq.test(table_financial_depression)
print(chi_result2)
table_family_depression<-table(data1$Family.History.of.Mental.Illness,data1$Depression)
print(table_family_depression)
addmargins(table_family_depression)
chi_result8<-chisq.test(table_family_depression)
print(chi_result8)
qqnorm(data1$CGPA[data1$Depression == 0]); qqline(data1$CGPA[data1$Depression == 0])
qqnorm(data1$CGPA[data1$Depression == 1]); qqline(data1$CGPA[data1$Depression == 1])
# Run Wilcoxon rank-sum test
wilcox.test(CGPA ~ Depression, data = data1)
# Median CGPA by depression group
aggregate(CGPA ~ depression, data = data1, FUN = median)
# Mean CGPA by depression group
aggregate(CGPA ~ depression, data = data1, FUN = mean)

# Create 3 age categories
```

```

data1$age_group <- cut(data1$Age,
                      breaks = c(-Inf, 20, 23, Inf),
                      labels = c("<=20", "21-23", ">23"))
table_age_depression <- table(data1$age_group, data1$Depression)
print(table_age_depression)
# Extract counts of depressed students in each age group
x <- table_age_depression[, "1"]
# Total students in each age group
n <- rowSums(table_age_depression)
# Test for equality of proportions
prop.test(x, n)
prop_depressed <- x / n
print(prop_depressed)
# Find the age group with highest proportion
names(prop_depressed)[which.max(prop_depressed)]

```

Logistic regression:

```

# Build logistic regression model
model <- glm(Depression ~ Gender + CGPA + Academic.Pressure + Sleep.Duration +
            Study.Satisfaction + Financial.Stress + Family.History.of.Mental.Illness +
            Have.you.ever.had.suicidal.thoughts + Age + Dietary.Habits + Work.Study.Hours, data = data, family =
            binomial)
summary(model)
library(car)
vif(model)
pred_probs <- predict(model, type = "response")
# Classify: 0.5 threshold
pred_class <- ifelse(pred_probs > 0.5, 1, 0)
# Confusion matrix
conf_matrix <- table(Predicted = pred_class, Actual = data1$Depression)
# Extract values
TN <- conf_matrix["0", "0"]
FP <- conf_matrix["1", "0"]
FN <- conf_matrix["0", "1"]
TP <- conf_matrix["1", "1"]
# TPR = Sensitivity = TP / (TP + FN)
TPR <- TP / (TP + FN)
# FPR = FP / (FP + TN)
FPR <- FP / (FP + TN)
# Print results
cat("TPR:", TPR, "\n")
cat("FPR:", FPR, "\n")
# Install (only once)
install.packages("pROC")
library(pROC)
# Predict probabilities
pred_probs <- predict(model, type = "response")
# Convert actual to numeric if needed
actual <- as.numeric(as.character(data1$Depression))
# Create ROC object
roc_obj <- roc(actual, pred_probs)
# Plot ROC curve
plot(roc_obj, col = "blue", main = "ROC Curve")
abline(a = 0, b = 1, lty = 2, col = "gray")
# Print AUC
auc(roc_obj)

```